



PSMA PET/CT and 11C-Choline PET/CT for Evaluating Disease Progression Following Lutetium-177–PSMA Therapy in Metastatic Castration-Resistant Prostate Cancer

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Introduction and Background

Prostate cancer is the most common cancer affecting men worldwide. Lutetium-177–PSMA (177Lu-PSMA) demonstrated a key role in management of advanced prostate cancer, especially in patients with castration-resistant disease who have failed hormonal and chemotherapy treatments. In light of studies reporting low PSMA expression and tumor heterogeneity in patients undergoing treatment with Lutetium which raises concerns on using only PSMA PET/CT in monitoring patient’s response. Herein, we sought to assess post Lutetium-177-PSMA radiographic progression using PSMA PET and Choline PET scans.

Materials and Methods

Using the Mayo Clinic Advanced Prostate Cancer Registry, we reviewed 512 PET/CT scans of 235 patients with mCRPC treated with 177Lu-PSMA between 2020 and 2025. All patients underwent baseline PSMA PET/CT prior to therapy. The post treatment scans comprised 195 11C-Choline PET/CT scans and 317 PSMA PET/CT scans. Radiographic progression was confirmed by conventional imaging, biopsy or response to subsequent treatment. A comparative descriptive analysis was performed to assess differences in progression detection between tracers

Results

Table 1 summarizes clinicopathological variables. Median age, Gleason score, prior treatments, and baseline disease distribution were similar between both groups. Among 195 11C-Choline PET/CT scans, 106 (54%) demonstrated radiographic progression, compared with 128 (40%) of 317 PSMA PET/CT scans. Median PSA at radiographic progression was 2.0 ng/mL for 11C-Choline PET/CT and 4.1 ng/mL for PSMA PET/CT (p = 0.003).

Conclusion

In patients treated with 177Lu-PSMA, 11C-Choline PET/CT detected radiographic progression more frequently and at lower PSA levels compared with PSMA PET/CT. These findings highlight potential differences in tracer performance following therapy and support the consideration of a multimodal imaging approach for post–177Lu-PSMA disease assessment. Further prospective studies are warranted to validate these results.



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	C-11 Choline SCAN (106)	PSMA PET SCAN (128)	P value
Age at Diagnosis of PCa, median (IQR), years	71.09 (66.643-76.052)	71.64 (66.836-77.340)	0.3135
Primary Gleason Score, median (IQR)	7 (8 - 9)	7 (8 - 9)	0.56
Median of PSA (IQR) at DX	11.365 (5.6-37.8)	12.3 (5.675-50)	0.81
Primary Treatment of PCa, n (%)			
RP	55 (51.89%)	67(52.34%)	0.945
RT	17 (16%)	18.75 (21.65%)	0.566
Chemo hormonal	28 (26.42 %)	26 (20.31%)	0.271
Unknown	7 (6.6%)	11(8.594 %)	
Distribution of Mets disease before Lutetium			
local	34 (32.08%)	31 (24.22%)	0.182
Lymph Node	67 (63.21%)	76 (59.38%)	0.550
Bone	83 (78.30%)	109 (85.16%)	0.1738
viscera	17 (16.04%)	28 (21.88%)	0.259
Median (IQR) of Lutetium cycles	6(6-6)	6 (5-6)	0.0003
Median (IQR) of PSA at radiographic disease progression After Lutetium therapy	2.1 (0.155-10.3)	4 (0.87-29.195)	0.0034